

STUDY OF PLASMID-MEDIATED ANTIBIOTIC RESISTANCE IN CLINICAL ISOLATES

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ABSTRACT:

The antibiotic resistance patterns of the seventy five clinical isolates were determined against five antibiotics including streptomycin, gentamicin, tetracycline, ampicillin, and vibramycin.

The Gram negative isolates were found resistant to many antibiotics as compared to the Gram positive clinical isolates. Vibramycin was found to be the most effective against 90% of the Gram negative and all the Gram positive clinical isolates. Other antibiotics were found to have moderate to poor effect against the clinical isolates.

The resistance of the clinical isolates was due to the presence of antibiotic resistant plasmid. This was determined by agarose gel electrophoresis. Resistant plasmid were studied for their curing by acridine orange. It was found that 66% of the Gram negative and 64% Gram positive clinical isolates have plasmid and the higher resistance pattern was due to the presence of R-plasmid.

INTRODUCTION

It is an established fact that indiscriminate use of antibiotics is followed by the development of resistant strains (Watanabe 1963). Antibiotics are not only used in medicine but also used in agriculture, food preservation, veterinary medicine and other fields (Best 1985). Prior to the antibiotic era resistant bacteria were rare. But the situation has changed with the use of antibiotics in clinical medicine and as animal additives. The resistance to antibiotic may be intrinsic or acquired (Davies & Smith 1978). The intrinsic resistance may be because of inadequate concentration of antibiotic action, or decreased permeability of the cell membrane to the antibiotics. The relative abundance of resistant organisms in a population is an increasing process as the antibiotic therapy continue. The continuing presence of antibiotic exert a selective pressure in favour of the resistant organism. Most of the resistance that occur clinically is due to the transfer of DNA fragment or plasmid from one organism to other via transformation, transduction or by conjugation (Lacey 1975).

Transferable R-plasmid are detected by conjugation whereas the existence of non-transferable R-plasmid is determined by their loss from the host spontaneously

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(Derylo & Lorkieweiz 1970). Such loss are increased when cells are treated with certain chemical agents such as acridine orange or ethidium bromide (Iyer & Iyer 1969). This phenomenon is referred as curing and has been used to ascertain the plasmid associated nature of genes (Michel-Briand & Laporte 1985; Watanabe & Lyang 1962).

The present research work deals with the study of several clinical isolates for their resistance to commonly used antibiotic and to find out whether the resistance is plasmid- mediated or not.

MATERIALS AND METHODS

The clinical isolates were obtained from various laboratories (Dr. Essa's Lab., Abbasi Shaheed Hospital etc.) and were identified by studying their growth pattern on different media and biochemical reactions. These culture were maintained on Tryptone Soya agar (Oxoid) by stabbing and stored at 4°C.

The antibiotic resistant pattern of clinical isolates to various antibiotics including streptomycin, gentamicin, tetracycline, vibramycin, and ampicillin was first determined by replica plate technique and then by tube dilution method (Ahmad *et al.*, 1987; Ahmad 1991).

Master plates of the isolated colonies from each culture were prepared (25 spots per plate). The impression were transferred by replica device to the plates containing varying concentration of different antibiotics. Plates were incubated overnight at 37°C and the results were recorded. The MICs of each isolate against different antibiotic were determined. MIC (minimum inhibitory concentration) is the lowest concentration of the antibiotic which inhibits visible growth of the organism. The MIC was also confirmed by broth dilution method using varying concentration of antibiotics.

The presence of plasmids were determined by minigel Agarose gel electrophoresis. Samples were prepared in lysing buffer and loaded in the slots of 1% Agarose submerged gel. The gels were run at 75 volts current for 1-2 hours. Bands were observed under transilluminator and results were recorded.

Ampicillin resistant isolates were randomly picked up for the curing of plasmid by acridine orange treatment. Fresh culture was prepared in tryptone soya broth. Acridine orange was added to get the final concentration of 50, 100 and 150 µg/ml and incubated at 37°C for 24 hours with control lacking acridine orange. After incubation the cultures were streaked on TSA plates to get isolated colonies and incubated at

37°C for 24 hours. After incubation 100 colonies from each dilution were grided onto TSA. After overnight incubation at 37°C impressions were transferred on to the plates containing 50 µg/ml of ampicillin (Clinical isolates were resistant to this concentration) by replica plate technique to check for loss of plasmid.

RESULTS AND DISCUSSION

Seventy five strains were isolated from clinical specimen supplied by various laboratories. The MIC was determined by replica plate technique and broth dilution method. Presence of plasmid was determined by Agarose gel electrophoresis and curing was carried out by acridine orange treatment.

The isolated cultures (50 Gram negative and 25 Gram positive isolates) were tested for their resistant pattern to five commonly used antibiotics in varying concentrations ranging from 1-500 µg/ml. The minimum inhibitory concentration of various antibiotics required to inhibit the growth of 90% and 50% isolates have been shown in Table-I. The MIC of both Gram negative and Gram positive clinical isolates ranged between 1- > 500 µg/ml of streptomycin, gentamicin, and ampicillin. Only vibramycin was found most effective antibiotic against both groups of micro-organisms and gentamicin and tetracycline comparatively less. Generally Gram positive bacteria were found to be more sensitive to the antibiotics used then the Gram negative bacteria Table-I.

Agarose gel electrophoresis was performed to determine the presence of resistant plasmid in all the clinical isolates. More than 50% bacteria were found to contain plasmid. Among Gram negative isolates about 66% were found to contain plasmid whereas 64% of the Gram positive were found to contain plasmid (Table-II).

Curing was performed to ascertain that the antibiotic resistance of the clinical isolates was due to the presence of resistant plasmid. Nine clinical isolates were treated with acridine orange. When *S. typhi* I was treated with 50 µg/ml of acridine orange, it was observed that 15% of the spots of *S. typhi* I (from isolated colonies) became sensitive to ampicillin, indicating the loss of resistant plasmid. Out of nine R-plasmid studied, five were lost by acridine orange treatment. Some of the R-plasmid incurable with acridine orange were found to be stable. Results with other organisms are shown in Table-III.

The stability of R-plasmid after chemical treatment reflects that these are very dangerous from epidemiological point as the disease caused by resistant plasmid (transferable by conjugation) containing organisms will be difficult to control (Ansari & Khatoon 1996).

The high incidence of antibiotic resistance among clinical isolates has helped in selection and dissemination of transferable antibiotics resistance among pathogenic bacteria. Overcrowding in cities and hospitalisation of patients provide ideal conditions for dissemination for resistant bacteria and indiscriminate use of antibiotics results in selective survival of resistant strains (Sydney *et al.*, 1980). The data shows that ampicillin had MIC₅₀ of 400 µg/ml against Gram negative bacteria and 100 µg/ml against Gram positive bacteria (Table-I). These concentrations are much higher than those attained in serum during therapy (Davis *et al.*, 1978).

Table I
Minimum Inhibitory Concentration (MIC) of Clinical Isolates
to various antibiotics

Antibiotics	MIC ₅₀ (in µg/ml)		MIC ₉₀ (in µg/ml)		MIC Range (in µg/ml)	
	G -ve	G +ve	G -ve	G +ve	G -ve	G +ve
Ampicillin	400	100	500	400	1- > 500	1- > 500
Gentamicin	50	1	400	300	1- > 500	1- > 500
Streptomycin	400	200	> 500	> 500	1- > 500	1- > 500
Tetracycline	100	50	400	100	1- > 500	1- 200
Vibramycin	5	1	50	25	1- 100	1- 50

Table II
Presence of Plasmids in Clinical Isolates

Bacteria	Plasmid present	% of Bacteria
Gram negative (50)	33	66%
Gram negative (25)	16	64%
Total Isolates (75)	49	65%

Table III
Curing of some randomly selected clinical isolates by
Acridine Orange treatment

Culture	Percentage of Curing after treatment with different concentration of Acridine Orange		
	50 µg/ml	100 µg/ml	150 µg/ml
<i>S. typhi</i> I	15	--	--
<i>Klebsiella</i> sp.	--	--	--
<i>E. coli</i>	--	13	10
<i>S. typhi</i> II	--	--	--
<i>Proteus</i>	--	--	--
<i>Pseudomonas aeruginosa</i>	--	--	2
<i>Citrobacter</i> sp.	--	--	--
<i>Staphylococcus aureus</i>	5	1	--
<i>Streptococcus</i> sp.	3	1	--

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